

Ujaan Das

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EDUCATION

University of Warwick <i>Master of Science in Computer Science (Predicted Distinction)</i>	Coventry, United Kingdom Sep. 2025 – Sep. 2026
Hong Kong University of Science and Technology <i>Bachelor of Engineering in Computer Engineering</i>	Hong Kong SAR Sep. 2021 – May 2025
Northwestern University <i>Exchange Semester</i>	Illinois, USA Jan. 2024 – Apr. 2024
West Island School <i>International Baccalaureate (41/45; HL: Computer Science, Physics, Psychology)</i> <i>iGCSEs (English (A*), Mathematics (8), Sciences (9); Art (7), Business Studies (B), Computer Science (A), Spanish (A), Global Perspectives (A))</i>	Hong Kong SAR Sep. 2019 – May 2021 Sep. 2017 – May 2019

EXPERIENCE

Stellerus Technology <i>Software Engineering Intern (Part Time)</i> - Re-designed a chatbot execution pipeline in Python using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session - Built a reusable React chat widget with multi-session support, enabling concurrent chats on platform - Integrated into SpaceGPT and awarded a <u>Gold Medal</u> at the 50th Geneva International Invention Exhibition	Jan. 2025 – Aug. 2025 Hong Kong SAR
Research Assistant <i>CASTLE Lab, HKUST</i> - Built a scalable Java static-analysis platform using Spoon to analyze JUnit test suites across 30k+ repositories - Developed graph-based analysis pipelines to model Java code structure, dependency relationships, control flow, and API interactions across multi-million-line codebases, generating minimally runnable unit test contexts - Automated build workflows for Maven and Gradle projects using Nix, generating over 35k usable examples	May 2024 – Sep. 2024 Hong Kong SAR
Stellerus Technology <i>Software Engineering Intern</i> - Developed RESTful APIs using FastAPI and MongoDB for multi-user task queues for live satellite data queries - Designed to sustain throughput of 8,000+ requests/sec with sub-150ms latency and minimal queue times - Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34% - Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%	May 2023 – Aug. 2023 Hong Kong SAR
HCI Initiative, HKUST <i>Research Assistant</i> - Developed a React/TypeScript SPA utilizing Custom Hooks and Context API for centralized state handling - Engineered a real-time telemetry and metrics pipeline processing 50K+ events/session, enabling analysis across 4K+ user interactions in large-scale experiments - Worked with other post-docs to build a standardized TailwindCSS and MaterialUI component library - Contributed to FARPLS, a feature-augmented preference learning system (published at ACM/IUT'24)	Jun. 2022 – Aug. 2022 Hong Kong SAR

PROJECTS

MSc Dissertation - Invariants C++, Google Test Suite, CMake, Python - Applying formal systems theory to LLMs for improved structured output generation via runtime constraints - Architecting a C++ DSL and compiler with Python bindings for seamless integration into inference pipelines	Github
Snip C++, Termios, Google Test Suite, CMake, Nix Actively maintaining a modal text editor built from ground-up with asynchronous, event-driven architecture	Github
yumcha Go, C, CGo, Cocoa, Nix Engineered a macOS tiling window manager in Go and C using Cocoa/CoreGraphics, implementing an i3-style tree layout engine for dynamic, scriptable window management without disabling SIP	Github
Undergraduate FYP - Follow-Me Robot C++, Python, ROS2, Raspberry Pi, Docker	Github

- Developed a real-time UWB tracking system via custom ROS2 Python/C++ nodes, applying multi-rate Kalman and EMA filtering for stable sub-centimeter localization under noisy conditions

oojsite | *Go, HTML, TailwindCSS, Nix*

[Github](#)

- Built a static site generator in Go to compile Markdown into composable HTML, actively used for [my blog](#)

TerraViz | *Next.js, React, TypeScript, CesiumJS, TailwindCSS*

[DevPost](#)

- Built an interactive geospatial platform in Next.js/React for real-time exploration of climate and land-use data

SKILLS

Languages: C++, Python, Go, C, TypeScript, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: Google Test Suite, CMake, FastAPI, Next.js, React.js, Node.js, LangGraph, Pydantic

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis

Tools & Cloud: Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git